Prediction for Economists

Prediction as an objective

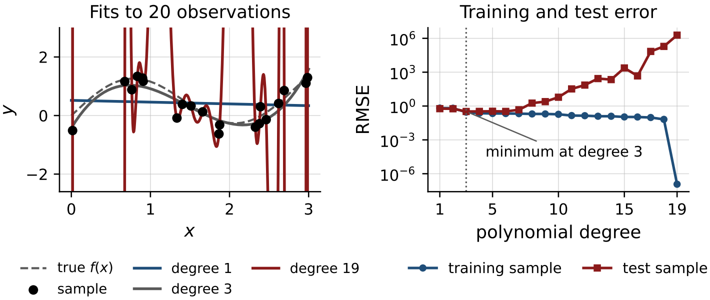
- A rule: observed characteristics in, an unobserved outcome out.
- Estimated on one sample, judged on another.
- The criterion is error on the second, not the value of any coefficient.
- Bias in the coefficients is acceptable when it lowers that error.

Everything that follows is about estimating that error honestly.

Outline

1. Overfitting and the trade-off
2. Estimating out-of-sample error
3. Leakage
4. Measures of accuracy
5. Constructing regressors
6. Choosing an estimator
7. Putting it together

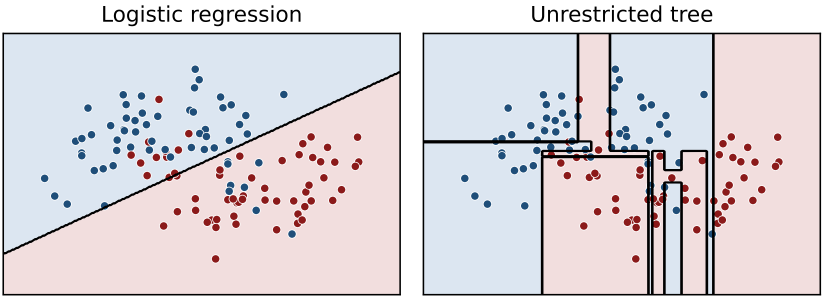
Overfitting



- In-sample fit: better at every degree.
- 20 observations, degree 19: every point exactly.
- Error in an independent sample: lowest at a low degree.

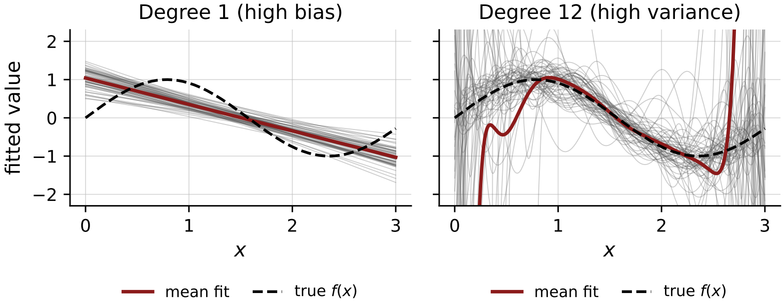
Fit in the estimation sample is not evidence of accuracy elsewhere.

Overfitting with a binary outcome



- Overfitting is not a property of polynomials or of continuous outcomes.
- Left unrestricted, the tree keeps splitting until each region is pure.
- It ends up isolating single observations, and misclassifies new ones.

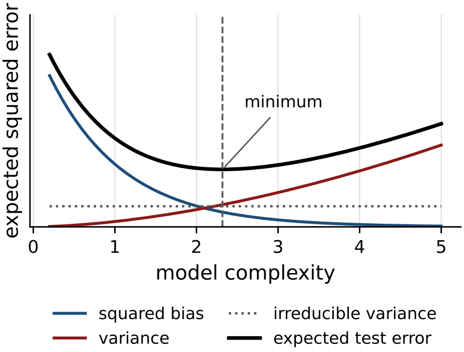
Bias and variance



- Each grey line: one independent sample of 25 observations.
- Restrictive fit: stable across samples, wrong on average.
- Flexible fit: right on average, far off in any one sample.

Two distinct errors, and less of one buys more of the other.

The bias–variance trade-off



- Squared bias, variance, noise.
- Restrictive: bias dominates.
- Flexible: variance dominates.
- The noise is irreducible.
- One point on the curve is best.

Which point, though? The curve is drawn here because the truth is known.

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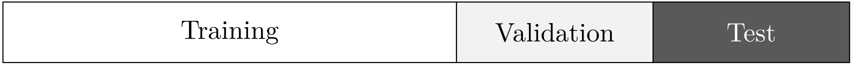
Holding observations back



- Out-of-sample error is unobserved. Hide observations and imitate it.
- Training: estimate the coefficients of one given specification.
- Validation: measure error there to *choose* — which regressors, which estimator, how flexible.
- Test: measure error once more, to *report* what the chosen rule will do.

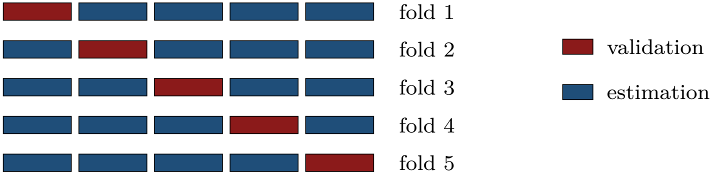
Both measure error; they answer different questions.

Why the test sample is separate

- Compare twenty specifications on the validation sample. One wins.
- It won partly on merit, partly because that sample's noise suited it.
- So the winner's validation error is better than its true accuracy.
- The more specifications compared, the larger that gap.

The test sample took part in no comparison, so its error is not flattered by having been chosen on. That is the number to report.

Cross-validation



- One validation block wastes those observations, and is one arbitrary draw.
- Instead, cut the training data into k folds of equal size.
- Each fold takes the validation role in turn; the other $k-1$ estimate.
- Average the k errors: one number per specification, using all the data.

The validation role rotates inside the training data; nothing is permanently set aside. The test sample stays outside the rotation, still unused.

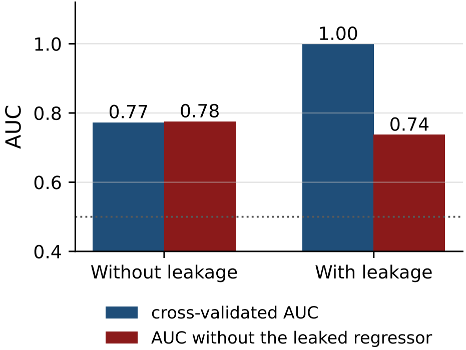
Outline

What leakage is

- Leakage: the fit uses information that will not be available when the rule is actually applied.
- The estimate of accuracy is then an estimate of a task that is easier than the real one.
- It shows up as a good cross-validated error and a disappointing one in use.

Nothing in the procedure complains. The fold was held out, so the arithmetic is correct; it is the data that were contaminated.

Leakage in a cross-validated fit



- Here one regressor was built from the outcome itself.
- Cross-validation reports an almost perfect fit.
- Remove that regressor and the fit is ordinary.

The held-out fold was held out, but its outcome had already travelled into the regressors.

Two kinds of leakage

- Wrong date: a regressor recorded after the outcome, or unknown at the moment the prediction would be made.
- Wrong sample: a quantity computed on the whole data and then used inside the folds.
- The second is the common one: standardization, imputation, category means, the choice of which regressors to keep.

Both make the cross-validated error too low, and the error in use a surprise.

Keeping the estimate honest

- Split first, transform afterwards. Never the reverse.
- Means, standard deviations and imputed values are estimated on the training fold, then applied to the held-out one.
- A regressor enters only if it is known before the outcome is realized.
- The test sample is opened once, at the end.

In code, this is why the transformations and the estimator are fitted together, inside the fold, rather than beforehand.

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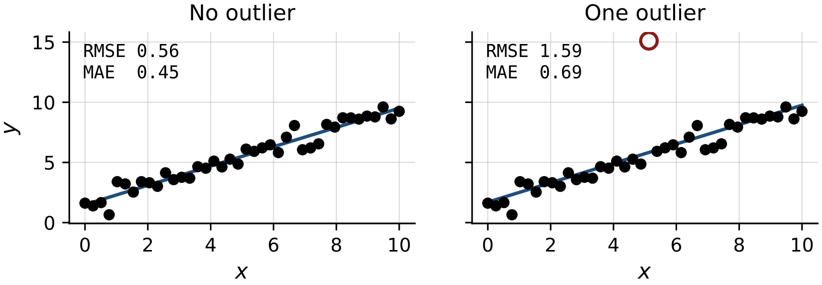
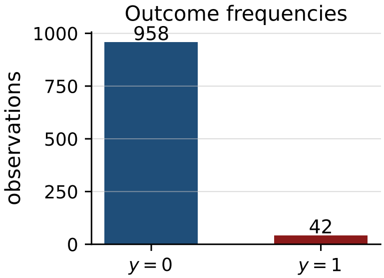
Measuring accuracy: continuous outcomes

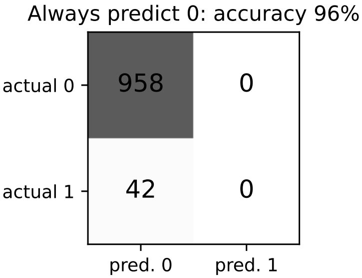


- Root mean squared error: large errors count for more.
- Mean absolute error: every unit of error counts the same.
- A few extreme observations move the first, not the second.

Fix the measure before the estimation, by the cost of an error.

Measuring accuracy: binary outcomes





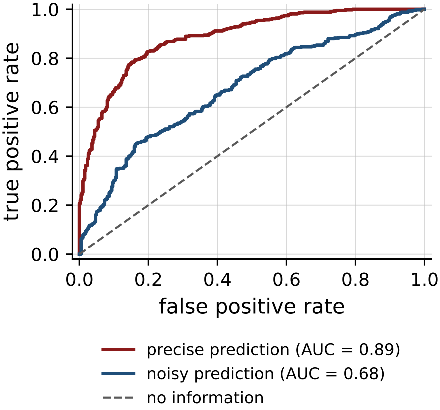
- Rare ones: the share of correct classifications says little.
- Zero everywhere: 96% correct, nothing identified.
- Predicted against actual, as a table: the useful summary.

Thresholds, precision, and recall

- Estimators return a probability. A classification needs a threshold.
- Precision: of those predicted one, the share that are one.
- Recall: of those that are one, the share predicted one.
- A higher threshold: more precision, less recall.

The threshold is a decision about the cost of the two errors, not an estimation problem.

The ROC curve



- Both rates, at every threshold.
- Its area: the quality of the ordering.
- No one threshold involved.
- The diagonal: no information.

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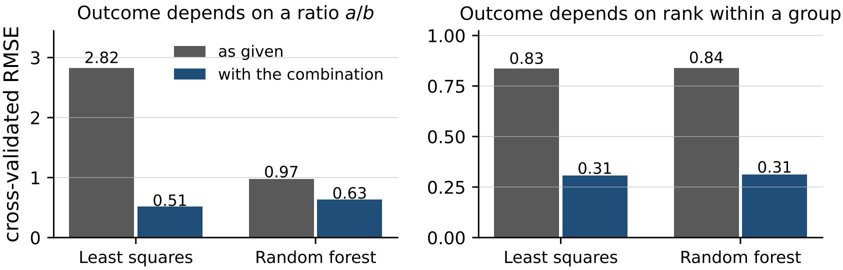
An estimator adds or splits; it never combines



- A linear model is additive; a tree is a step function on rectangles.
- Neither forms a ratio, a group mean, or a rank within a group.
- Both hold the ingredients here, and fifteen irrelevant regressors besides.
- Handing over the combination is what moves each of them.

Which combinations

- Ratios: debt to assets, price to earnings, output per worker.
- Deflated, real and per-capita terms, so units compare across time and size.
- Differences and growth rates: the level and the change are different regressors.
- Position within a group: size against the industry median, income against the region, a rank within the cohort.

An economist can write these down in a morning. No estimator will discover them in any amount of data.

Three standard constructions

- Principal components: many correlated regressors replaced by a few. Chosen without reference to the outcome, so the first need not predict.
- Lags, differences and rolling means, where the data have a time dimension — each dated to exist at the moment of the prediction.
- The outcome itself: fitting to $\log y$ penalizes relative error rather than absolute.

Did the regressor earn its place?

- Shuffle one regressor, leave the rest alone, and see how much the cross-validated error worsens.
- Nothing worsens: the regressor carries nothing, whatever its coefficient.
- Almost everything worsens under one regressor: the signature of leakage.
- Any estimator, and it reports on the regressor rather than on a coefficient.

This is what the constructions of the last three slides are to be put through.

Outline

Three decisions

- Which family: penalized linear regression, or something tree-based.
- How flexible within that family: the penalty, the depth, the number of trees.
- When to stop comparing candidates, and which one to report.

Least squares is the first entry, as the benchmark the rest have to beat.

All three are settled by the cross-validated error

- What we want is error on observations not used in the estimation. Nothing else is the objective.
- Fit in the estimation sample cannot settle anything: it improves with flexibility whatever the truth is.
- Nor can theory. Which family wins depends on the shape of the relation and on how much noise there is, and both are unknown.
- Cross-validation estimates the objective itself, so the comparison is made in the units that matter.

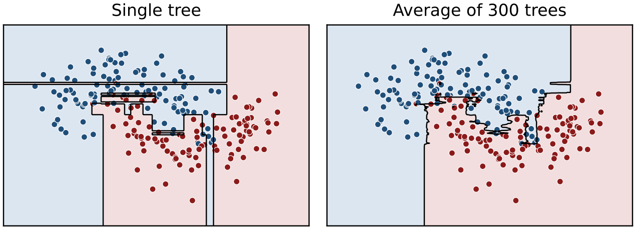
Same regressors, same folds, one number for each candidate. That number decides the family, the penalty, the depth, and the number of trees alike.

When to stop

- That number is an average over k folds, so it carries a standard error of its own.
- A gap between two candidates smaller than that standard error is not evidence that either is better.
- Every further candidate compared raises the chance that the winner won on the noise of these particular folds.
- So take the simplest candidate within one standard error of the best, and stop there.

The one-standard-error rule; Hastie, Tibshirani and Friedman, sec. 7.10.1.

The random forest as a default



- One tree: the boundary is a feature of the particular sample.
- A forest averages many trees on resampled data. Most of the variance goes.
- Two choices, and little turns on either. Nothing to scale or transform.
- One estimator for a continuous outcome and for a binary one.

Fernández-Delgado et al. (2014) compared 179 classifiers from 17 families over 121 data sets. The random forests came out best.

Outline

The workflow

1. Split → 2. Construct regressors → 3. Estimate → 4. Cross-validate → 5. Select → 6. Evaluate

- Was every transformation fitted inside the fold?
- Could any regressor contain the outcome?
- Same regressors and same folds across every specification?

Step 6 happens once. Anything learned there is not a reason to return to step 2.

Summary

- Accuracy concerns observations not used in the estimation.
- In-sample fit does not measure it; bias against variance is the trade-off.
- Cross-validation estimates it. Leakage quietly destroys the estimate.
- Fix the measure of accuracy in advance, by the cost of an error.
- Regressors first, estimator second.

Appendix

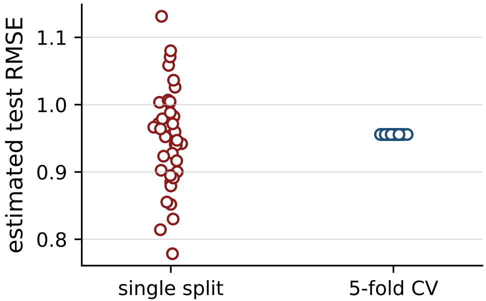
The decomposition of expected prediction error

For a new observation at x_0, with $y = f(x_0) + \varepsilon$ and $\hat{f}$ estimated on a random sample,

$$\mathbb{E}\Big[(y - \hat{f}(x_0))^2\Big] = \big(\mathbb{E}[\hat{f}(x_0)] - f(x_0)\big)^2 + \mathrm{Var}\big(\hat{f}(x_0)\big) + \mathrm{Var}(\varepsilon).$$

- The expectation is taken over samples, not over observations.
- The first two terms can be traded against one another.
- The third cannot be reduced by any estimator.
- Penalization and averaging lower the second at some cost in the first.

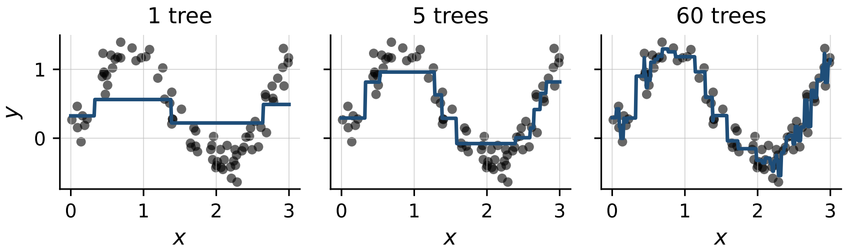
Sampling variability of a single split



- Each point: a new partition.
- One split varies a great deal.
- The average over folds is stable.
- Small samples: it matters most.

With a binary outcome, keep the share of ones equal across folds.

Boosting: successive fits



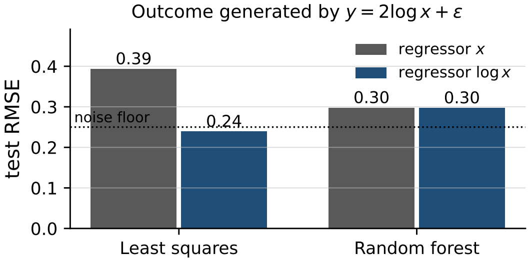
- A single tree of depth two is a coarse step function.
- Successive trees fitted to the residuals accumulate into an accurate fit.
- Continued far enough, the procedure begins to fit the noise.
- The number of trees is itself chosen by cross-validation.

Bet on sparsity

- If many regressors each matter a little, no procedure predicts well: the data cannot support that many estimated effects.
- If a few matter a lot, penalized methods find them.
- So act as though few matter. Build the candidates, let the penalty select.
- The bet is not always right. It is that there is nothing better to bet on.

Hastie, Tibshirani and Friedman, sec. 16.2.2. Stated there as a principle, not a theorem.

Transformations of a single regressor



- A tree splits on $x \leq t$, so it reads the ordering of x and nothing else.
- An increasing transformation leaves its partition, and its fit, unchanged.
- The forest scores 0.30 either way; the same log is worth 0.15 to least squares.

Rescaling one regressor: everything to a linear model, exactly nothing to a tree.
Combining two regressors: worth doing for both.

Categorical variables: the encoding is a decision

- Few levels: one indicator each, and nothing further is needed.
- Many levels: as many regressors as levels, most of them nearly empty, and each estimated on a handful of observations.
- Group the rare levels together, or replace a level by its frequency. The frequency uses no outcome, so it can be computed before the split.
- Replacing a level by its mean outcome predicts well and leaks: the outcome enters the regressor. Compute it within the fold or not at all.

Missing values carry information

- Values are rarely missing at random: no credit history, a firm too young to report, a question refused.
- Impute the value, and add an indicator for having imputed it.
- The indicator is often the better predictor of the two.
- Both the imputation and the indicator are built inside the fold.

Dropping the incomplete observations throws that information away, and leaves a sample selected on the regressors.

Further measures for binary outcomes

- When ones are rare, the precision–recall curve is more informative.
- ROC can look favourable when few of the predicted ones are correct.
- The F_1 measure combines precision and recall in one number.
- Log-loss penalizes confident predictions that turn out to be wrong.
- Calibration: of those given probability 0.7, about 70% should be ones.

References

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- James, Witten, Hastie and Tibshirani, *Introduction to Statistical Learning*.
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